

Revolutionizing Connectivity: Visible Light Communication for Wireless Transmission

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Abstract—Visible Light Communication (VLC) can be concisely stated as Data Transmission Through Illumination. Data transmission through visible light communication is much more secure and is capable of achieving high data transmission rates as compared to existing conventional wireless technologies like Wi-Fi, Bluetooth, Wi-max, etc., which uses radio frequency spectrum. This project aims at building a wireless VLC system capable of transmitting text data, sound and image between two computers using visible light. A LASER is used as the transmitter, air as the transmission medium and a Solar Cell as the receiving component. Text, sound and image data gets transmitted as strings of 1s and 0s as the LASER flickers on and off, rapidly at a rate undetectable to the human eye. This binary data is received by the Solar Cell, converted into the appropriate format and displayed on the output screen. Both hardware (Arduino Uno microcontroller) and software (Arduino IDE) are required for successful data transmission and reception. Two computers are used as the end devices.

Keywords: Visible Light Communication, wireless technologies, LASER, Solar Cell.

1. INTRODUCTION

VLC (Visible Light Communication) refers to transmission of information using visible light portion (380nm to 780nm) of the electromagnetic spectrum. LASER is used as a transmitter, air as the transmitting medium and an appropriate Solar Cell acts as the signal receiving component. In the last two decades, there has been a tremendous increase in the demand for wireless data communication. Wireless communication using radio frequency spectrum being the most popular till the present era has led to the congestion of this spectrum [1]. The advent of Visible light communication opens up a new possibility for future communication between any kind of devices. Firstly, visible light spectrum bandwidth, which ranges from 430 THz to 750 THz is much larger than the radio frequency spectrum bandwidth ranging from 3 KHz to 300 GHz. Due to this larger bandwidth, it is possible to achieve a much higher rate of data transfer and also a large number of users can be accommodated [2]. Secondly, using radio frequency can be life threatening in certain places like space stations, hospitals and mines whereas visible light being a sustainable means of existence of life is unquestionably safe [3]. Thirdly, light sources being ubiquitous and abundant, once replaced with LASERS or LED lamps, can also act as transmitters in addition to acting as light sources [4]. This ensures the abundance of visible light communication systems and the low cost to implement them. Lastly, since light cannot propagate through walls unlike radio waves, the data cannot be received by a receiver present on the other side of the wall [5]. This property of light ensures security, as information

here, is not broadcasted for external sources to receive. This technology is still in its early stage of development In 2006 a research work proposed a broadband communication system for indoor applications as well [6]. In 2010 a research work has been able to establish a visible light communication system which can transmit data up to 500 Mbit/s but this technology is still several years away from proper commercialization. LASER, in addition to their illumination capabilities, require low power for their operation and have very high switching speeds such that intensity variations of LASER are so rapid that they are undetectable to the human eye. The purpose of this research work is to establish a visible communication system using Arduino microcontroller. In this paper a simple system is proposed which uses two Arduino UNO boards, Red LASER as transmitter) and Solar Cell as receiver. Arduino IDE is used for data processing at both transmitting and receiving ends. In this system, the user can enter up to 50 characters in the transmitting end. Arduino causes the LED to flicker at a higher rate which is undetectable to the human eye. It changes the intensity of light from a higher range to a lower range depicting pulses of 1s and 0s respectively. In the receiving end the Solar Cell receives the transmitted data as a bit stream which is then converted back into characters. Arduino IDE is used for taking character input from the user in the transmitter end and for converting the binary bit stream back into character form in the receiver end. Similarly in transmission of sound these binary bits are fed to the PAM module connected with a speaker.

2. SYSTEM MODEL

A system model (as shown in fig. 1.), is designed and constructed. The system consists of 2 PCs, one at the transmitter end and the other at the receiver end, two microcontrollers (Arduino UNO) at both ends, a red LASER at the transmitter end and a Solar Cell at the receiver end.

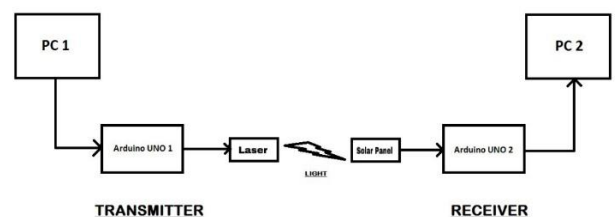


Fig. 1.

Block Diagram for data transmission between two PCs using Visible Light Communication

At the transmitting end, the PC is connected to the Arduino UNO which is connected to the red LASER. The Arduino causes the LASER to flicker rapidly at a speed undetectable to the human eye [7], transmitting data simultaneously. These rapid flickering of LASER creates pulses with different durations [8]. At the receiving end the Solar Cell is connected to the Arduino which is connected to the PC. The Solar Cell detects the rapid flickering of LASER. By using a special function in the Arduino IDE, the duration of the pulses, created by the flickering, are calculated. Depending on the durations, HIGH and LOW pulses are distinguished and a binary bit stream is generated. This bit stream is further processed and output is generated.

A. TRANSMITTER

In the transmitter side of the VLC system a personal computer (PC 1) is used. The Arduino IDE software needs to be installed in PC 1 in order to facilitate the execution of the system. This computer acts as the data source. An Arduino UNO board (Arduino UNO 1) is connected to PC 1 via a USB cable. A red LASER is connected to digital pin number 7 of the Arduino board. Arduino programming is done in order to take an input from the user and transmit the respective input data [9]. An interfacing is done between Arduino IDE and Arduino to facilitate this transfer of input data. Firstly, a serial port object (s) associated with serial port COM6 of the Arduino is created. The Baud Rate that is the number of signal or symbol changes occurring per second is set to 9600 bits per second. The serial port object is then connected to the Arduino. The Arduino IDE program takes an input from the user in the form of a string and sends this input to the Arduino. The serial port object is then disconnected from the Arduino. A computer code written in Arduino IDE software, receives the string input from the user either statically or dynamically. It converts the string into its respective ASCII value and then generates its corresponding binary bit stream. Firstly, digital pin number 7 on the Arduino board is defined and configured to behave as an output. Then the baud rate is set to 9600 bits per second for serial communication. The number of bits (characters) available for reading from the serial port is obtained and the incoming serial data is read. A HIGH or a LOW value is written to the digital pin 7. Pulse Width Modulation (PWM) is implemented here that is the duration of the pulse is varied according to the bit stream [10]. For example- if the transmitting string is 101, one high pulse will be transmitted for bit 1 followed by a low pulse for bit 0 followed by another high pulse for the least significant bit 1. A specific delay is provided for high pulse=1. Similarly, another specific delay is provided for low pulse=0. Thus, data gets transmitted as the LASER flickers rapidly according to the different duration of delay provided for the high pulse and the low pulse [11]. Human eye detects frames every 1/10th of a second. Thus, the rapid flickering of the LASER is undetectable to the human eye. Similarly, while transmission of Audio signal, the audio is converted in the strings of 1s and 0s which are transmitted through the LASER. These strings are detected by Solar Cell which convert this light signal into weak electric current. These weak electric current are converted into audio signals using a speaker acting as a transducer and the audio is further amplified with the help of

PAM module that employs the technique of Pulse Amplitude Modulation.

Arduino IDE code for transmitter is shown below:

```
#define LASERPIN 7
void setup() {
  // put your setup code here, to run once:
  pinMode (LASERPIN, OUTPUT) ;

  char myText[] = "A quick brown fox jumped over the lazy dog";
  int length = sizeof(myText);
  int ar[50];
  int m;
  int bits[8];
  for (int i =0 ; i<length ; i++ ) {
    ar[i]= int(myText[i]);
  }
  for (int n =0 ; n<length ; n++){
    m=ar[n];
    int z;
    int bin[7];
    int newbin[7];
    for(z=0;z<8;z++){
      bin[z] = m%2;
      m = m /2 ;
    }

    for (int j= 7 ; j>=0 ; j-- ){
      newbin[7-j] = bin[j] ;
    }
    for( int p=0 ; p<8 ; p++ ){
      if (newbin[p] == 1){
        bits[p] = HIGH ;
      }
      if ( newbin[p] == 0){
        bits[p] = LOW ;
      }
    }
  }
  bits[0] = HIGH;
```

```
for (int i = 0; i < 8; i++) {
  digitalWrite(LASERPIN, bits[i]);
  delay(10);
}

digitalWrite(LASERPIN, LOW);

delay(100);
}
```

B. RECEIVER

An Arduino UNO board is connected to the PC 2 via a USB cable. A Solar Cell is connected to the analog pin A13 of the Arduino UNO 2. The Solar Cell is placed 1 meter away from the LASER such that a direct line of sight is maintained between them. The Solar Cell detects the rapid flickering of the LASER since its output varies with the variations in the amount of light incident on it from the LASER, while transmitting 0 or 1[12]. A predefined library

function `pulseIn()` is used to read the incoming (HIGH or LOW) pulses to the A13 pin. This function returns the width or durations of the pulses in microseconds. Two ranges of time durations are obtained with the help of `pulseIn()`. Now as specific delays or specific pulse durations have been set for high and low pulse at the transmitter end, the LASER glows with a certain intensity when high pulse is transmitted and with a different intensity when low pulse is transmitted. The Solar Cell is capable of sensing this intensity variations.

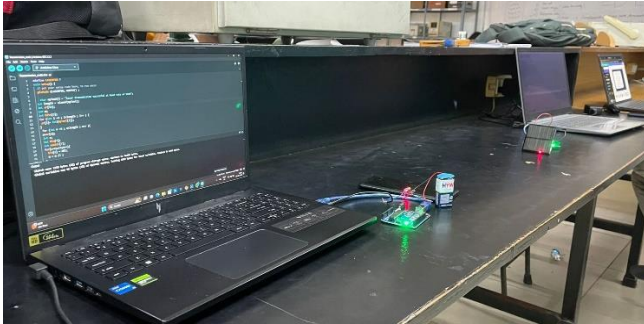


Fig. 2.

Hardware Circuitry of Transmitter and Receiver

After the duration of the received pulses are obtained with the help of `pulseIn()`, the high and the low pulses are easily detected with the help of a conditional if else statement where pulses having longer duration are treated as high pulse=1 and pulses having shorter duration are treated as low pulse=0. After obtaining the high and the low pulses, a binary bit stream of 0 and 1 is generated. The obtained bit stream is then processed by Arduino IDE for obtaining an output in the form of a string, identical to the input provided by the user. At the receiving end, a bit stream is received by Arduino. Characters are extracted from this received bit stream. Each character is represented by 8 bits. But redundant bits are present at times. For example, while transmitting 1, 2, 3 characters, 8, 16, 24...bits are to be considered respectively at the receiving end. A loop is initiated and the entire stream of bits is stored in a vector form. Then the required number of bits are extracted corresponding to the number of characters transmitted. The transmitted bit stream gets stored in a single column at the receiving end. In order to obtain the transmitted stream of bits in a single one dimensional array, a predefined library function `reshape()` is used. The bit stream is then converted to character array (String) with the help of the function `char()`. A second loop is initiated which goes on for n cycles where n=number of characters given as input. At the end of each cycle, 8 bits are taken i.e. for the first cycle it will take up to 8 bits in a string. In the second cycle, 9th to 16th bits are considered in a string and in the third cycle, it will take 17th to 24th bits and so on. But each time the characters need to be retrieved and stored. So, in each cycle 8bits are stored in an array. Then the array is converted into a string by using the predefined library function `num2str()`. There is yet another problem to be dealt with. The transmitted bit stream is received in reverse order. Suppose the transmitting bit stream is 100 then it will be received as 001 at the

receiving end. So the received bit stream needs to be reversed in order to get a correct output. The string is reversed by using the predefined `flipr()` function. Now the required 8 bits are suited for character conversion. Their respective ASCII values are obtained by operating a predefined library function `bin2dec()`. Predefined library function `char()` takes the ASCII values and generates the respective characters. The characters are stored in an empty character array. At the end of the loop all characters get stored in a string. The string is displayed using the function `disp()` after exiting the loop. Thus, the received data displayed on the screen of PC 2 is identical to the string user input provided for transmission.

Arduino IDE code for receiver is shown below:

```
#define SOLARPIN A0
#define THRESHOLD 210

int ambientReading = 0;

void setup() {
    pinMode(SOLARPIN, INPUT);
    Serial.begin(9600);
}

void loop() {

    int reading = analogRead(SOLARPIN);

    int bits[8];

    //Listening for the start bit
    if (reading > THRESHOLD) {
        for (int i = 0; i < 8; i++) {
            if (analogRead(SOLARPIN) > THRESHOLD) {
                bits[i] = 1;
            }
            else {
                bits[i] = 0;
            }
            delay(10);
        }

        int m = 0;
        for (int j = 1; j < 8; j++) {
            if (bits[j] == 1) {
                m = m + (1 << (7 - j));
            }
        }
    }
}
```

```

}

char n=m;
Serial.print(n);

}

```

In the receiving end the Solar Cell senses the pulses and depending on the length of the pulses it creates the binary bit stream. Arduino IDE reads this binary data and processes it. Now after receiving the binary data a lot of processing still needs to be done. The whole bit stream is in reverse order. Each character is represented by 8 bits. The bit stream needs to be divided and arranged properly, all these functions are achieved by Arduino IDE software.

3. RESULT

A model was designed and successfully constructed in order to demonstrate Serial communication between two PCs. The objective was to establish a visible light communication system which transmits and receives text. In the initial stage, Arduino UNO microcontroller and Arduino IDE were used to transmit and receive a single character. Later on upto 50 characters were successfully transmitted and received. The distance between the LASER and the Solar Cell was set to 1 meter. A pictorial representation of a Pulse Width Modulated (PWM) signal, followed by pictorial representations of the Arduino and Arduino IDE depicting the transmission and reception of data, is included in the following page.

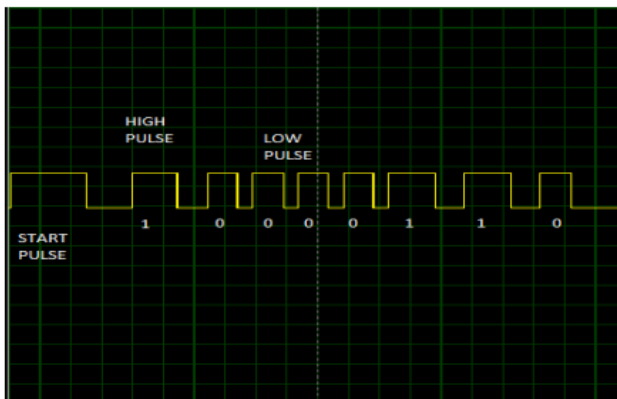


Fig. 3.

Generated pulses while transmitting a single character

Figure 3 shows the generated pulses for one character. Binary '1' is represented by a HIGH pulse and binary '0' is represented by a LOW pulse.

```

Transmission_code | Arduino IDE 2.3.2
File Edit Sketch Tools Help

Arduino Uno Upload

Transmission_code.ino
1 #define LASERPIN 7
2 void setup() {
3   // put your setup code here, to run once:
4   pinMode (LASERPIN, OUTPUT);
5
6   char myText[] = "laser transmission succesful at baud rate of 9600";
7   int length = sizeof(myText);
8   int ar[50];
9   int m;
10  int bits[8];
11  for (int i =0 ; i<length ; i++) {
12    ar[i]= int(myText[i]);
13  }
14  for (int n =0 ; n<length ; n++){
15    m=ar[n];
16    int z;
17    int bin[7];
18    int newbin[7];
19    for(z=0;z<8;z++){
20      bin[z] = m%2;
21      m = m /2 ;

```

Fig. 4.

Transmitting a string of characters using VLC

Figure 4 shows the console of Arduino IDE at the transmitting end. After running the code the console takes a string as user input which in this case is 'laser transmission succesful at baud rate of 9600'.

```

sketch_mar1a | Arduino IDE 2.3.2
File Edit Sketch Tools Help

Arduino Uno

sketch_mar1a.ino
20 if (reading > THRESHOLD) {
21   for (int i =0; i<8 ; i++) {
22     if (analogRead(SOLARPIN) > THRESHOLD) {
23       bits[i] = 1 ;
24     }
25     else {
26       bits[i] = 0;
27     }
28     delay(10);
29   }
30
31   int m = 0;
32   for (int j =1; j <8; j++) {
33     if (bits[j] ==1) {
34       m = m + (1<<(7-j));
35     }
36   }
37
38   char n=m;
39   Serial.print(n);
40
41 }
42 }

Output Serial Monitor x
Message (Enter to send message to 'Arduino Uno' on 'COM7')
laser transmission succesful at baud rate of 9600

```

Fig. 5.

Receiving a string of characters in Arduino IDE

Figure 5 shows the console of Arduino IDE in the receiving end. After running the code the console shows a string as the output which is same as the input in the receiving end.

As for the case of transmission of sound, the sound converted into strings of 1s and 0s by LASER is detected by Solar Cell, which convert this light signal into weak electric current. These weak electric current are converted into audio signals using a speaker acting as a transducer and the audio is further amplified with the help of PAM module that employs the technique of Pulse Amplitude Modulation.



Fig. 6.

Output model for reception of sound

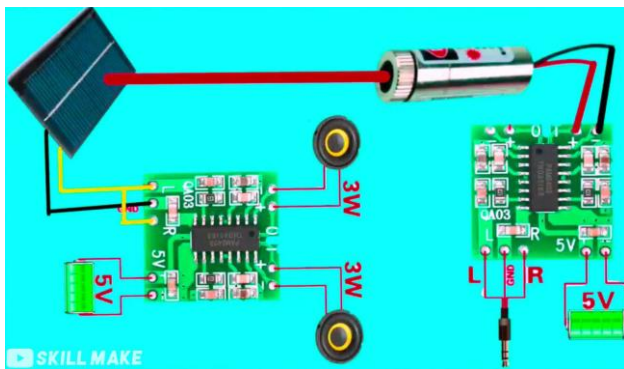


Fig. 7.

Output circuitry for reception of sound

4. CONCLUSION

In this paper a visible light communication system between two personal computers using Arduino microprocessors has been established. The system designed here is capable of transmitting and receiving upto 50 characters over a maximum range of 3 meters. The value of pulse durations may vary with

different lighting conditions thereby affecting the flickering of LASER in the presence of ambient light. Red LASER being used as the optical transmitter and Solar Cell as an optical receiver, cost of building such a system was considerably low.

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